

Postdoctoral Researcher (from Mar. 2026)
 Advanced Robot Control Lab (advisor: *Yisoo Lee*)
 Center for Humanoid Research,
 KIST (Korea Institute of Science and Technology)

seokin.seo@kist.re.kr
seokinseo11@gmail.com
seokinseo.github.io
 Seoul, South Korea

Education

KAIST, Ph.D. in Kim Jaechul Graduate School of AI, 2026.
 ► **Dissertation:** Robust Imitation Learning under Data Collection Biases
 (co-advised by *Kee-Eung Kim*, *Hongseok Yang*)
KAIST, M.S. in School of Computing, 2019. (advised by *Kee-Eung Kim*)
 ► **Thesis:** A Study on Generating Explanations for Reinforcement Learning
 Policies in Control Tasks
Sogang University, B.S. in Computer Science & Mathematics, 2017.

Research Interests

Reinforcement Learning, Imitation Learning, Physical AI
 Reliable sequential decision-making from imperfect data, toward physical robots.

Patents

**Regularized Behavior Cloning for Blocking the Leakage of
 Past Action Information**
 KR Patent Application, No. 10-2023-0159062, Nov. 2023.

Awards

Gold Reviewer Award
 International Conference on Machine Learning (ICML), Jul. 2026.
Outstanding Paper Award
 Korea Software Congress (KSC), Feb. 2023.

Academic Experiences

Guest Lecturer
 Mathematics for AI (AI503), KAIST (Fall 2025): *Distribution Shift in IL*
Teaching Assistant
 Introduction to RL, KAIST Machine Learning Engineer Bootcamp (2022)
 Mathematics for AI (AI503), KAIST (Fall 2021)
 Dialogue System, KAIST–Samsung DS AI Expert Program (2020)
 Machine Learning (CS376), KAIST (Fall 2019)
 Data Structure (CS206), KAIST (Spring/Fall 2018)
Academic Talk
 Robust and Safe Offline Imitation Learning via Distribution Matching,
 Locomotion and Manipulation Technology Workshop, Korea University (2026)
 Mitigating Covariate Shift in Behavioral Cloning via Robust Stationary
 Distribution Correction, AI Tutorial Series, KAIST (2025)
Academic Reviewer
 NeurIPS (2021, 2024, 2026), ICLR (2025, 2026), ICML (2025, 2026)

Programming Skills

Languages & Frameworks
 Python (PyTorch, JAX), C, C++, Ruby (Rails), Java, JavaScript
 Anaconda, Docker, Git, Kubernetes, Nomad
Simulator Experiences
 MuJoCo (MJX, MJWarp), IsaacGym, IsaacSim, CARLA

International Conference

- [I9] **Seokin Seo**, Sung Kuk Shyn, Minji Seo, Yisoo Lee[†], Jongmin Lee[†], Hongseok Yang[†], Kee-Eung Kim[†].
DexDICE: Reconciling Safety and Performance via Dual-Expert Offline Imitation Learning.
40th Conference on Neural Information Processing Systems (NeurIPS), 2026. **(To appear)**
- [I8] Yoonseok Choi, Chaeyoung Oh, Hyunjun Choi, **Seokin Seo**, Kee-Eung Kim.
Concept Removal Guidance: Evidence-Calibrated Negative Guidance for Safe Diffusion Sampling.
Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026. **(Spotlight)**
- [I7] Joo Bon Maeng*, Seongmin Lee*, **Seokin Seo**, Kee-Eung Kim.
Goal-Conditioned DPO: Prioritizing Safety in Misaligned Instructions.
Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL), 2025.
- [I6] **Seokin Seo**, Byung-Jun Lee, Jongmin Lee, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
Mitigating Covariate Shift in Behavioral Cloning via Robust Stationary Distribution Correction.
38th Conference on Neural Information Processing Systems (NeurIPS), 2024.
- [I5] **Seokin Seo**, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
Regularized Behavior Cloning for Blocking the Leakage of Past Action Information.
37th Conference on Neural Information Processing Systems (NeurIPS), 2023. **(Spotlight)**
- [I4] HyeongJoo Hwang, **Seokin Seo**, Youngsoo Jang, Sungyoon Kim, Geon-Hyeong Kim, Seunghoon Hong, Kee-Eung Kim.
Information-Theoretic State Space Model for Multi-View Reinforcement Learning.
Proceedings of the 40th International Conference on Machine Learning (ICML), 2023. **(Oral)**
- [I3] Geon-Hyeong Kim, **Seokin Seo**, Jongmin Lee, Wonseok Jeon, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
DemoDICE: Offline Imitation Learning with Supplementary Imperfect Demonstrations.
International Conference on Learning Representations (ICLR), 2022.
- [I2] Youngsoo Jang, **Seokin Seo**, Jongmin Lee, Kee-Eung Kim.
Monte-Carlo Planning and Learning with Language Action Value Estimates.
International Conference on Learning Representations (ICLR), 2021.
- [I1] Wonseok Jeon, **Seokin Seo**, Kee-Eung Kim.
A Bayesian Approach to Generative Adversarial Imitation Learning.
32nd Conference on Neural Information Processing Systems (NeurIPS), 2018. **(Spotlight)**

Preprints & Under Review

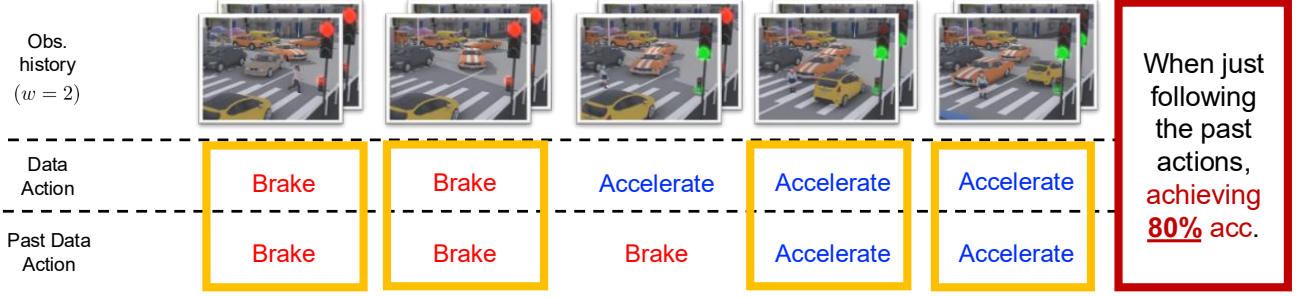
- [P2] Sol Choi, Mingon Kim, **Seokin Seo**, Jongeun Choi, Donghyeon Kim, Yisoo Lee.
Reusing Locomotion Training Recipes Across Humanoids via Body-Aligned Virtual Joints.
Under review in International Conference on Humanoid Robots (Humanoids), 2026.
- [P1] Seongmin Lee*, Jaewook Shin*, Young Jin Ahn, **Seokin Seo**, Ohjoon Kwon, Kee-Eung Kim.
Zero-Shot Multi-Hop Question Answering via Monte-Carlo Tree Search with Large Language Models.
arXiv preprint arXiv:2409.19382, 2024.

Domestic Conference & Journals

- [D2] **Seokin Seo**, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
Improving Causally-Regularized Logistic Regression via Confounder Balancing with Feature Combinations.
Korea Software Congress (KSC), 2022. **(Outstanding Paper Award)**
- [D1] Geon-Hyeong Kim, **Seokin Seo**, Shinhyung Kim, Kee-Eung Kim.
Variational Recurrent Neural Networks with Relational Memory Core Architectures.
Journal of KIISE, Vol. 47, No. 2, pp. 189–194, 2020.

Selected Publications

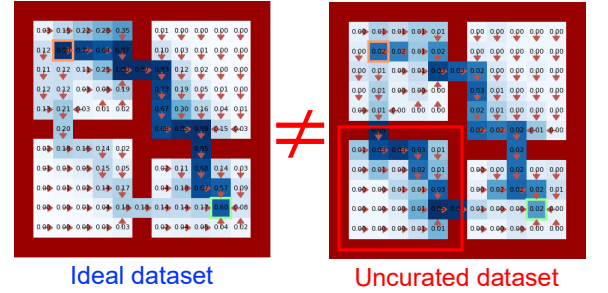
1. PALR: Mitigating Action Leakage in History-Based IL (NeurIPS 2023, Spotlight) [I5]



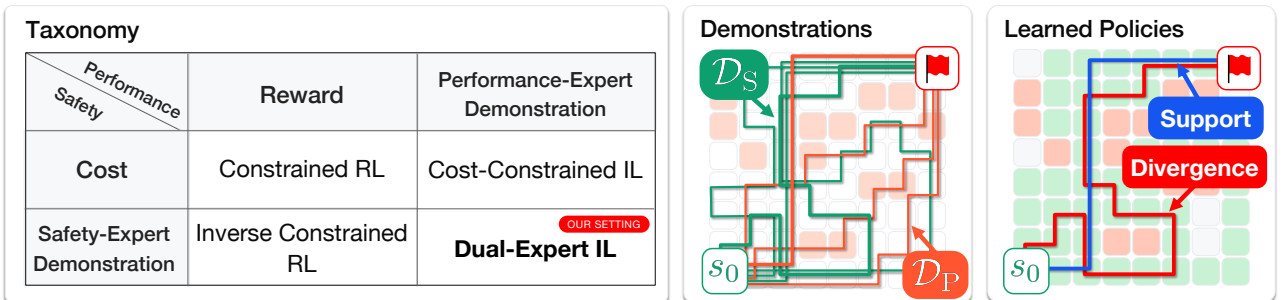
- We study imitation learning from observation histories, which are commonly used in partially observable settings (e.g., vision-based control) to capture control-relevant quantities such as velocity and acceleration.
- In this setting, offline IL often induces a strong dependence on past actions; we (i) quantify this dependence and (ii) propose a regularized behavioral cloning approach to mitigate it.
- We evaluate across diverse simulators, including state-based robotics tasks in MuJoCo and vision-based tasks in the CARLA autonomous driving simulator.

2. DrilDICE: Robust Offline IL under Covariate-Shifted Expert Datasets (NeurIPS 2024) [I6]

- Prior IL methods often assume that the dataset coincides with the expert policy’s stationary distribution. However, this assumption is frequently violated in realistic data-collection settings (For example, when trajectories are segmented due to long horizons or when collection bias is present.) even if the action labels are of expert quality.
- We develop an offline IL method that is robust to **covariate shift** by (i) formulating the problem as distributionally robust optimization (DRO) and (ii) deriving a practical algorithm, DrilDICE, from an offline RL interpretation of the resulting objective.



3. DexDICE: Safe Offline IL with Dual-Expert Demonstrations (NeurIPS 2026) [I9]



- Prior safe IL methods typically assume access to a cost function, which is often difficult to specify. Instead, we leverage two distinct sets of expert demonstrations: a performance-relevant set $\mathcal{D}_P$ and a safety-relevant set $\mathcal{D}_S$.
- In the offline setting, we optimize a DICE (distribution correction estimation)-based offline IL objective that constrains the agent to remain close to $\mathcal{D}_P$ while staying within the support of $\mathcal{D}_S$.
- We evaluate our method on extensive benchmarks, including (i) a recurated dataset for safe RL (DSRL) and (ii) a real-world mobile navigation task with obstacle avoidance, under a range of constraints.